

TECHNICAL

PYTHON	SQL
PYTORCH	XGBOOST
POLARS	PANDAS
FASTAPI	OPENCV
CLAUDE AGENT SDK / MCP	
DOCKER	GCP

CORE COMPETENCIES

- Monte Carlo simulation and Bayesian state estimation for live markets.
- Calibration-first evaluation vs. devigged closing lines — CRPS, conformal intervals.
- CLV grading, walk-forward backtesting, Kelly sizing.
- Autonomous multi-agent LLM fleets for research and execution.

ACADEMIC TRACK

B.S. DATA SCIENCE

University of Iowa
2022 — PRESENT
3 YRS COURSEWORK · CONTINUING ONLINE

CERTIFICATION

CLAUDE CODE IN ACTION	ANTHROPIC
MODEL CONTEXT PROTOCOL	ANTHROPIC
CDS™	USDSI
CBIAP	IABAC

LEADERSHIP

EAGLE SCOUT — BSA

Led 20+ service projects and a 12-person team; 500+ volunteer hours.

TENNIS TEAM CAPTAIN

Regional lead; ran training and player development.

KESEM / KIDS FOR CAUSE

Support network for children of cancer patients.

PROFESSIONAL PROFILE

Quantitative researcher and agentic engineer building autonomous prediction systems for live sports markets — simulation engines, calibration-first ML pipelines, and multi-agent LLM fleets that run research, execution, and grading end-to-end, benchmarked against the devigged closing line.

SELECTED RESEARCH & DEVELOPMENT

COURT VISION

PRIMARY QUANT RESEARCH PLATFORM

LIVE RESEARCH

CV-MTCA-V8

CV tracking → possession-level Monte Carlo → **live pricing across 372 correlated markets** — run 24/7 by an autonomous multi-agent AI fleet.

- One sim, every market:** a single joint distribution prices spreads, totals, and props — 372 markets per game.
- CV tracking:** YOLOv8 + ByteTrack + court homography feed spatial features (xPPP) into the models.
- CLV-graded execution:** every entry graded on closing-line value, not raw ROI, with off-market price guards.
- In-play edge:** live engine conditions pregame priors on realized game state, where pricing is softest.
- Honest evaluation:** leak-free walk-forward backtests — 2,300+ predictions graded vs. devigged closes.
- Agentic ops:** Claude agent fleet self-heals 18+ microservices and runs overnight research loops.

MULTI-SPORT KERNEL

Sport-blind kernel — NBA, MLB, and soccer share one calibration, backtest, and execution harness.

AGENTIC RESEARCH FLEET

Claude Agent SDK + MCP agents with persistent memory; changes ship only if they beat the benchmark.

SPATIAL INTELLIGENCE

KDE shot heatmaps, K-Means player archotyping, homography court-space projection.

REVENUE FORECASTING

Prophet demand forecaster (<8% MAPE) with dbt + BigQuery pipelines on GCP.

PROFESSIONAL EXPERIENCE

DATA ANNOTATION SPECIALIST

DATA ANNOTATION TECH | REMOTE

SPRING 2026

- Labeled 10,000+ samples for ML training and LLM fine-tuning.
- Wrote edge-case documentation that cut labeling errors ~20%.

DATA ANALYST INTERN

SUNSOLOR | TEMPE, AZ

SUMMER 2025

- Built Tableau / Power BI dashboards that cut manual reporting time 30%.
- Automated daily market-trend reports in Python; tuned SQL on 1M+ row data.

BUSINESS INTELLIGENCE ANALYST

FORTREX SECURITIES | MADISON, WI

2023 — 2024

- Built executive reporting interfaces with 99.9% uptime.
- Added real-time anomaly alerts to transaction-processing pipelines.